

UPAY PRESENTS · AI DEV FEST 2026 · CPC DIU

CYGNUS AI

Suspicious transaction network detection for upay
and Bangladesh's mobile financial services

Mapping money trails like constellations.

Team Sadekunnahar Marufa · Al Amin · Tanvir Hossain Shihab

Cygnus AI at a glance

01



The problem

Why mule networks
beat rule-based checks

02



Our solution

Graph patterns and
explainable AI

03



Live demo

Follow the Money, AI
Investigator, case flow

04



Results

Precision, recall, F1
and PR-AUC

05



Impact for upay

Savings, pilot plan,
roadmap

Tip: every slide has an [◀Agenda](#) link at the top right to come back here.

Every criterion, built, measured and shown.

CRITERION		WHAT CYGNUS AI DELIVERS	SLIDE
Problem relevance	20%	Public Bangladesh MFS figures, plus two upay detectors: risk-area account takeover and agent hundi	05 ▶
AI/ML depth	20%	8 models compared on 423k held-out transactions: best F1 0.89, PR-AUC 0.94, false positives 0.12%	16 ▶
Business impact	20%	73.6% fewer alerts at equal recall, false alarms 904 → 46, and an 8-week upay pilot to measure real savings	21 ▶
Prototype quality	15%	Working analyst flow: sign in, alert, trace, investigate, open case, escalate, STR/SAR export, audit trail	13 ▶
Innovation	10%	Rules miss 686 of 1,010 suspicious accounts; the graph model catches 561 of those	17 ▶
Scalability	10%	1.06M transactions scored in 18.9 s on one laptop process; API p95 under 6 ms; streaming + graph-DB path	19 ▶
Responsible AI	5%	Two roles, hash-chained audit log, PII masking with logged reveal, bias check by account tier and district	20 ▶

Fraud hides in the network, not the transaction



Each transfer looks normal

Mule networks split money so every single payment passes rule-based checks.



Investigations take hours

Analysts trace multi-hop flows account by account in flat tables.



Alerts without reasons

Black-box scores and threshold alerts are hard to defend to auditors.

WHAT RULES SEE

A-214 BDT 9,800



A-377 BDT 9,650



A-108 BDT 9,900



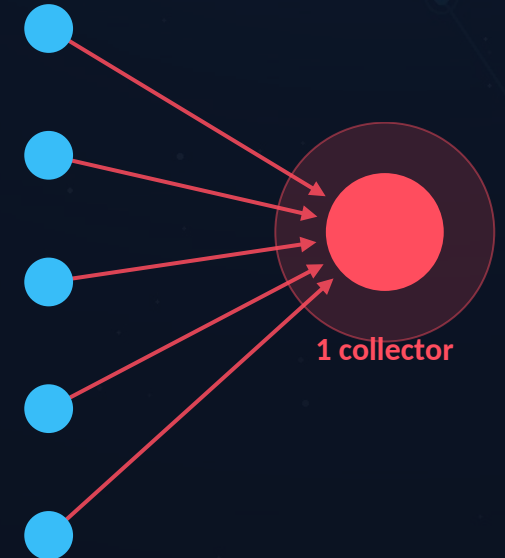
A-652 BDT 9,700



A-590 BDT 9,850



WHAT THE GRAPH SEES



Five 'clean' transfers are really one **fan-in collector** account.

Bangladesh MFS: huge volume, real laundering

239.3M

MFS accounts

January 2025 [1]

~22M

MFS transfers a day

678.6M in Oct 2025, worth Tk 1.58 trillion [2]

30,199

STRs / SARs to BFIU

FY2024-25, up 74% in one year [3]

Tk 75,000 cr

moved by digital hundi

in one year, via ~5,000 MFS agents (bKash, Nagad, upay, Rocket) [4]

BANGLADESH TYPOLOGIES IN OUR SYNTHETIC DATA

Digital hundi via agents: an agent wallet funded every month pays the same local wallets

Gambling collection: hundreds of small deposits into a few collector wallets

Mule rings: recruited wallets forward money within minutes and cash out

Structuring: repeated transfers just under a reporting threshold

Circular layering: money loops through 3 to 6 wallets back to its origin

Stolen-device takeover: a wallet suddenly used from an unfamiliar risk area on a new device

21,725 MFS accounts

suspended by BFIU in nine months of 2023 for online gambling and hundi [5].

**At ~22M transfers a day, nobody can read rows.
The network has to surface itself.**

[1] Bangladesh Business Standard Express, 23 Mar 2025 [2] Bangladesh Bank via Financial Express, 5 Jan 2025 [3] Biv via Financial Express, 15 Jul 2026 [4] CID via Business Standard, 8 Sep 2022 [5] BFIU via Business Standard, 27 Sep 2023

One mule ring, one trail across Bangladesh

1

Feeders pay in **FAN-IN + STRUCTURING**

Eight upay wallets send deposits just under BDT 10,000 to one hub wallet in Dhaka.

2

Moved on in minutes **RAPID MOVEMENT**

Within 1 to 3 minutes the hub forwards each deposit to four mule wallets.

3

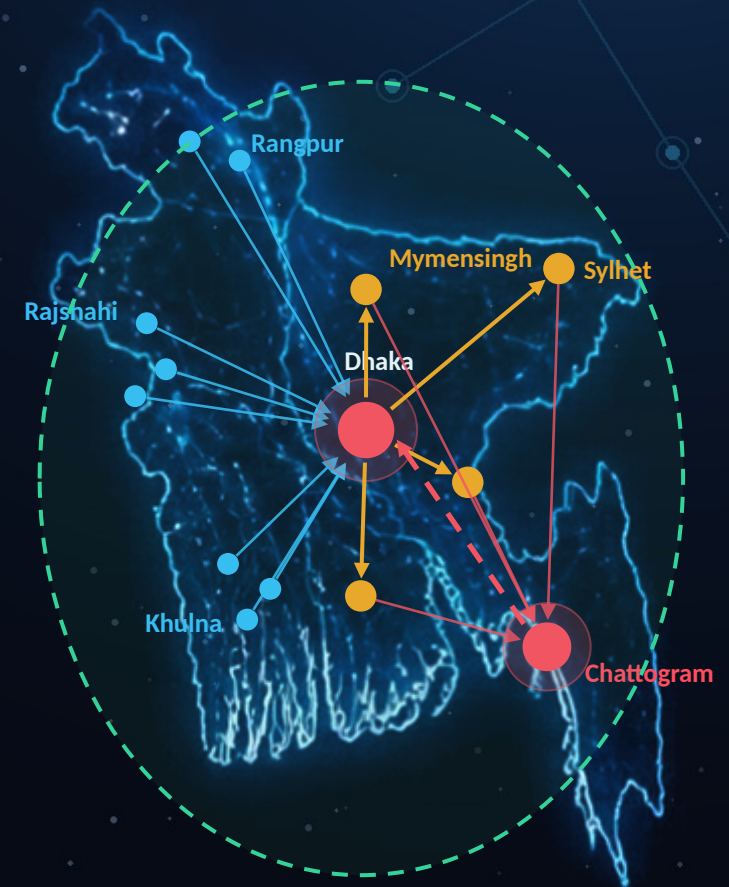
Cash-out and loop back **CIRCULAR FLOW**

The mules pay a cash-out agent in Chattogram, and 90% flows back to the hub.

4

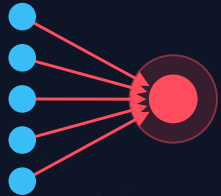
Cygnus AI flags the ring **92.3 CRITICAL**

Every transfer looked normal on its own. The graph scores the hub 92.3 CRITICAL, with evidence and next steps.



Click to reveal each step

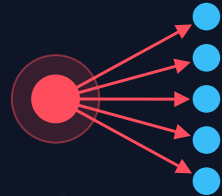
Seven suspicious shapes money leaves behind



3
accounts flagged

Fan-In

Many wallets funnel money into one collector within 2 hours.



2
accounts flagged

Fan-Out

One wallet sprays funds across many receivers in a burst.



13
accounts flagged

Rapid Movement

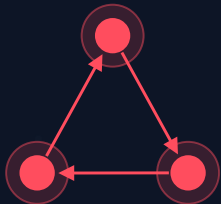
Incoming money is forwarded onward within minutes.



19
accounts flagged

Transaction Chain

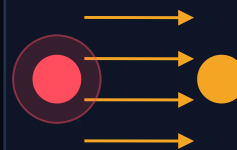
Funds hop through three or more intermediaries.



9
accounts flagged

Circular Flow

Money loops back to its origin within 6 hours.



4
accounts flagged

Structuring

3+ transfers just under the BDT 10,000 limit in 24 h.



4
accounts flagged

Coordinated Network

A dense cluster of 4+ accounts trading internally.



9
accounts flagged

ML Anomaly

Isolation Forest flags outliers, no fraud labels needed.

Location risk and agent hundi, built for upay



Risk-area account takeover

Risk judged against each customer's own baseline

- ▶ Learns each wallet's usual areas and handsets from its own history
- ▶ Risk areas come from upay's fraud team (demo uses placeholders Zone-13, Zone-14)
- ▶ Alert needs at least two signals: area new for this customer, new handset, unusually fast outflow, new receivers
- ▶ Also flags the collector wallet that gathers money from several taken-over accounts

No hassle for legit users Customers who regularly transact from a risk zone are not flagged. A first-time hit triggers step-up checks (PIN / OTP) or analyst review, never an automatic block.

Measured Precision 93.1%, recall 84.4%, false-positive rate 0.02%. Residents of risk areas flagged: 0 of 5,136. Visitors: 8 of 1,027. Collector wallets: 24 of 24.



Agent hundi operator

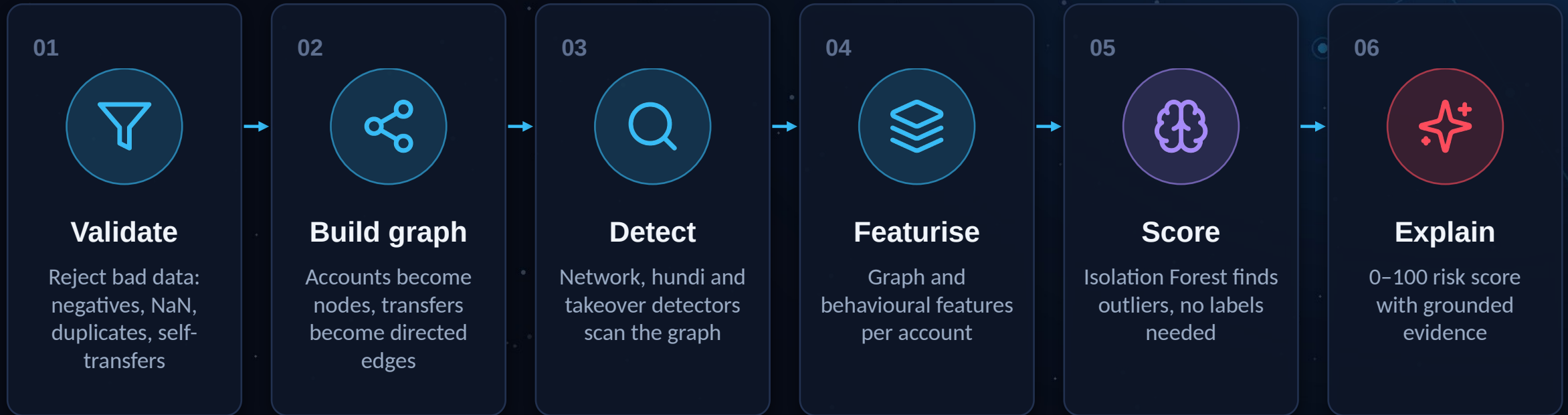
Recurring foreign-linked money through one agent

- ▶ Agent funded informally (no licensed inward-remittance or distributor source)
- ▶ Pays out to five or more beneficiaries soon after funding
- ▶ The same beneficiaries are paid again in two or more cycles
- ▶ Also flags the funders who keep topping the agent up

No hassle for legit users Agents doing licensed remittance payout are not flagged (0 of 117), and families who simply receive money are scored on their own history.

Measured Operator: precision 100%, recall 91.7%. Funders: 100% / 92.9%. Licensed remittance agents flagged: 0 of 117.

From raw transfers to an explained risk score



Accounts become nodes. Transactions become edges. **Suspicious shapes become evidence.**

Every risk score shows its working



$$\text{Risk} = 0.40 \times \text{ML} + 0.40 \times \text{Graph} + 0.20 \times \text{Behaviour}$$

WORKED EXAMPLE ACC_MULE_HUB · money-mule collection hub

ML anomaly		99.8
Graph pattern		100.0
Behaviour		62.0

92.3
CRITICAL RISK



Follow the Money, one hop at a time



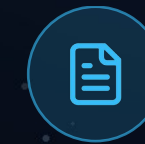
Pick a flagged account

Jump to any of 11 demo scenarios in one click.



Expand 1 or 2 hops

Directed edges show exactly where the money went next.



Read the evidence

Score, inflow, outflow and patterns for every account.

[Open live dashboard](#)

[Graph API docs](#)

[ML API docs](#)

From alert to next step in seconds



Names the typology

Money-mule hub, layering loop, smurfing: each case gets a plain-language label and summary.



Explains with real numbers

Key findings quote the pipeline's own metrics, so every sentence can be audited.



Recommends MFS next steps

Check NID, SIM and device links, review agent cash-outs, and for CRITICAL cases prepare an STR for BFIU.

Works fully offline. An optional Claude narrative rewrites the briefing using only the pipeline's facts.

AI Forensic Investigator

Every finding and next step below is built from this account's detected patterns, transaction metrics and Isolation Forest score.

CASE SUBJECT ENTITY

ACC_TKO_COLLECTOR

COMPOSITE SCORE

94.8 / 100

Account-takeover collector

RULE-BASED BRIEFING

ACC_TKO_COLLECTOR scores 94.8/100 (CRITICAL). It collects money from several wallets that were suddenly emptied from a risk area. Detected patterns: rapid pass-through, collection from suspected takeovers. Received BDT 50,964 and sent BDT 49,435 across 12 transactions.

KEY FINDINGS

- 01. Passed received money straight on 6 times (to 1 wallet), the fastest within 1 minutes of receipt.
- 02. Passed on 97% of everything it received, keeping almost nothing.
- 03. The Isolation Forest model ranks its overall behaviour as a statistical outlier (ML score 98/100).

Recommended Next Steps

- Hold outgoing transfers and cash-out on the collecting wallet under internal policy and review its KYC, device and agent linkage.
- List every wallet that paid it during the suspected takeovers and contact those customers.
- Review the timing of each receive-and-forward pair; funds held for minutes suggest the wallet is a relay, not the beneficiary.
- Check cash-out records for the destination wallets and the agents involved.
- Open linked cases for ACC_TKO_VICTIM_01, ACC_TKO_AGENT, ACC_TKO_VICTIM_02, ACC_TKO_VICTIM_03 in Follow the Money.
- Escalate to the compliance officer today; consider a temporary hold on outgoing transfers under internal policy and prepare a Suspicious Transaction Report (STR) for BFIU if the review confirms the pattern.

LINKED ACCOUNTS

ACC_TKO_VICTIM_01 ACC_TKO_AGENT ACC_TKO_VICTIM_02 ACC_TKO_VICTIM_03

Generated from synthetic data and computed metrics. These are risk indicators for prioritisation, not proof of wrongdoing.

High-Risk Accounts for Review

ACC_TKO_COLLECTOR

rapid_movement, takeover_collector

ACC_MULE_HUB

fast_in, circular_flow, transaction_chain, rapid_mov

ACC_MULE_01

circular_flow, transaction_chain, rapid_movement

ACC_MULE_02

circular_flow, transaction_chain, rapid_movement

ACC_MULE_03

circular_flow, transaction_chain, rapid_movement

ACC_MULE_04

circular_flow, transaction_chain, rapid_movement

ACC_HUNDI_AGENT

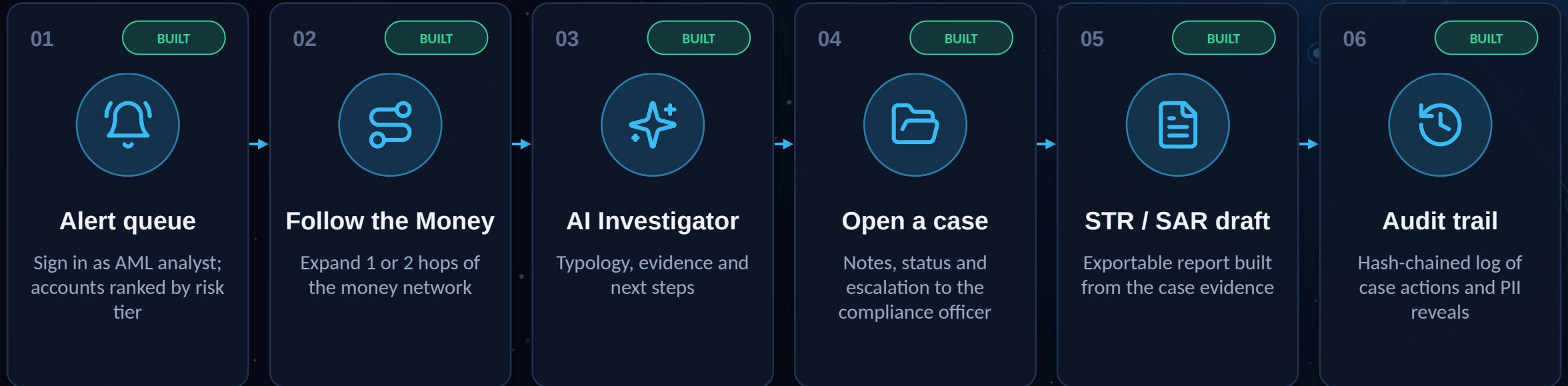
fast_out, fund_operator

CYGNUS AI

AI-Powered Transaction Network Intelligence

AI DEV FEST 2026 · SYNTHET

One flow, from alert to STR draft, all in the product



90-SECOND LIVE DEMO

Sign in as an AML analyst, open the top CRITICAL alert, follow the money two hops, read the AI Investigator, open a case and add a note, escalate it, export the STR/SAR draft, then show the audit log and the Model Evaluation tab.

[Open the live demo](#)

Runs on our laptop today. The public server gets this version once the PR is merged.

Cases, STR drafts and a tamper-evident audit log

Case Management

Alert, case, notes, escalation, decision, report. The model ranks and explains; a person decides, and the decision is kept as a label for retraining.

1

0

0

1

Refresh

Case moved to Escalated.

CASE QUEUE (1)

CASE-2026-0001EscalatedAccount takeover collector - ACC_TKO_COLLECTORP1 - CRITICAL 94.8 - analyst

CASE-2026-0001EscalatedCRITICAL

Account takeover collector - ACC_TKO_COLLECTOR

Opened 10/7/2026, 10:06:57 AM by analyst · priority P1 · assigned to analyst

RISK PICTURE AT CASE OPEN

Account takeover collector on account ACC_TKO_COLLECTOR

Score 94.8 / 100 - patterns: rapid movement, takeover collector

ACC_TKO_COLLECTOR scores 94.8/100 (CRITICAL). It collects money from several wallets that were suddenly emptied from a risk area. Detected patterns: rapid pass-through, collection from suspected takeovers. Received BDT 50,964 and sent BDT 49,435 across 12 transactions.

Trained classifier rates this account 99% similar to known laundering typologies. Main drivers: total received (BDT) = 509641 (typical 1164.65); share of transactions at 90-100% of the threshold = 0.833 (typical 0.1)

TIMELINE

Case opened10/7/2026, 10:06:57 AM · analyst (analyst)

Note added10/7/2026, 10:07:06 AM · analyst (analyst)

Status OPEN → ESCALATED10/7/2026, 10:07:07 AM · analyst (analyst)

View network →

Export STR draft

Case management Open, note, escalate to the compliance officer

Suspicious Transaction Report - Draft

Reference STR-DRAFT-CASE-2026-0001 · generated 2026-10-07T04:07:09.083Z by analyst (analyst)

CONFIDENTIAL - DRAFT FOR COMPLIANCE REVIEW. Synthetic demonstration data. Not submitted to any authority.

1. CASE

Case ID	CASE-2026-0001	Status	ESCALATED
Title	Account takeover collector - ACC_TKO_COLLECTOR		
Priority	P1	Assigned to	analyst
Opened	2026-10-07T04:06:57.742Z by analyst	Decision	Pending

STR / SAR draft Exported from the case, personal data masked

Model Evaluation

Synthetic Bangladesh-style MFS networks (personal wallets, agents, merchants) with six planted typologies and legitimate look-alike traffic. Models are trained on two networks, tuned on a third and tested on separate networks generated with different random seeds.

423,042test transactions
30,490 accounts, 1010 suspicious

0.943PR-AUC, graph model
rules only: 0.201

94.8%precision
rules only: 28.5%

83.8%recall
rules only: 32.1%

0.12%false-positive rate
rules only: 2.35%

74%fewer alerts at equal recall
graph model vs rules

Model comparison

Mean over the held-out networks. PR-AUC is the headline because suspicious accounts are rare; it does not depend on the alert threshold.

MODEL	TYPE	PR-AUC	PRECISION	RECALL	F1	FALSE-POSITIVE RATE	ALERTS / 1,000 ACCOUNTS	
Rules only (conventional monitoring)	rules		0.201	26.5%	32.1%	0.290	2.35%	31.1
Phase 1 Cygnus score	unsupervised		0.101	9.3%	21.1%	0.129	5.48%	58.0
Graph pattern detectors	rules on graph		0.488	22.1%	82.9%	0.348	7.69%	96.1

Model Evaluation tab The benchmark numbers, live in the dashboard

Audit Log

Append-only and hash-chained: every entry stores the SHA-256 hash of the previous one, so an edited or deleted line breaks the chain. No unmasked personal data is written here.

Chain intact: 8 entries verified
head: a35478ae104b2a83f6e8d43b5289ebcf9514b7f6441a8f0b3a4d4b6d93f652 · checked 10:07:22 AM

#	TIME	ACTOR	ROLE	ACTION	TARGET	DETAILS	HASH
8	10/7/2026, 10:07:18 AM	admin	admin	AUTH_LOGIN	-	method: demo	a35478ae104b...
7	10/7/2026, 10:07:09 AM	analyst	analyst	REPORT_EXPORTED	CASE-2026-0001	format: html - unmasked: false	3c735e39f15b...
6	10/7/2026, 10:07:07 AM	analyst	analyst	CASE_STATUS_CHANGED	CASE-2026-0001	from: OPEN to: ESCALATED	232940b77343...
5	10/7/2026, 10:07:06 AM	analyst	analyst	CASE_NOTE_ADDED	CASE-2026-0001	note_id: NT	d837cc72939d...
4	10/7/2026, 10:06:57 AM	analyst	analyst	CASE_OPENED	CASE-2026-0001	account_id: ACC_TKO_COLLECTOR · risk_level: CRITICAL · risk_score: 94.8	5b6099b20562...
3	10/7/2026, 10:06:34 AM	analyst	analyst	AUTH_LOGIN	-	method: demo	31a44c304555...
2	10/7/2026, 10:06:30 AM	admin	admin	PIPELINE_RUN	-	transactions: 239 · accounts: 89	50d703df93fb...

Audit log SHA-256 hash chain; one click verifies it

Every planted scenario, scored by the live product

239

transactions analysed

89

accounts mapped as a graph

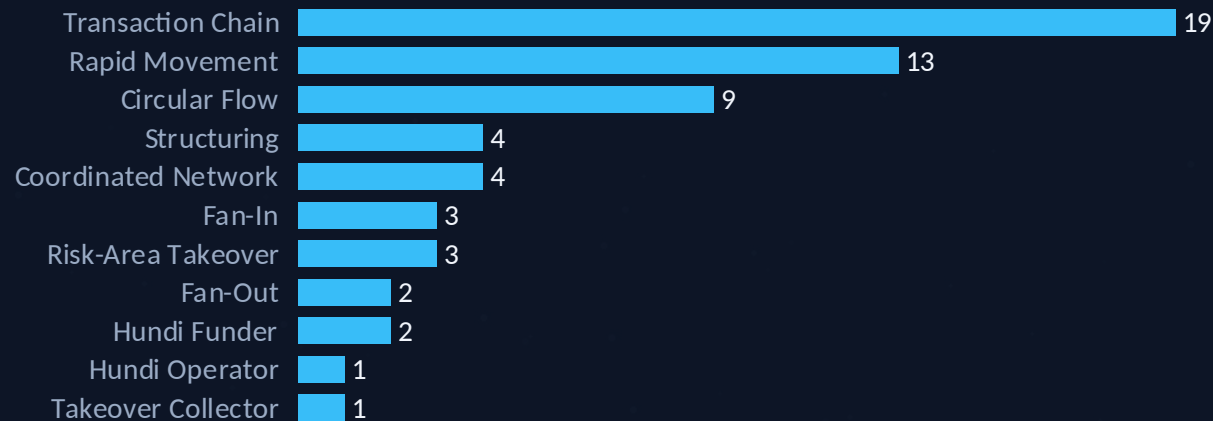
11 / 11

patterns detected

71 + 10

automated tests passing

Accounts flagged per pattern (demo dataset)



11 benchmark scenarios

Normal

Fan-In

Fan-Out

Rapid

Chain

Circular

Coordinated

Structuring

Mule ring

Hundi

Takeover

9 of the 10 planted fraud targets score MEDIUM or above; the rapid pass-through scores 28.3 (LOW) but is still flagged by its pattern. No normal account is flagged by any pattern.

Fixed random seeds: same input, same score.

Measured on 423,042 held-out transactions

423,042

labelled transactions

39,490

accounts

1,010

planted suspicious accounts

2 train · 1 tune · 3 test

evaluation split

Model	Precision	Recall	F1	PR-AUC	False-positive rate
Rules only (conventional monitoring)	26.5%	32.1%	0.29	0.20	2.35%
First prototype score (unsupervised)	9.3%	21.1%	0.13	0.10	5.40%
Isolation Forest (unsupervised)	8.2%	9.6%	0.09	0.09	2.83%
Graph pattern detectors	22.1%	82.9%	0.35	0.49	7.69%
Cygnus composite (current product)	69.8%	88.0%	0.78	0.84	1.00%
Gradient boosting, no graph features	80.3%	63.4%	0.71	0.76	0.41%
Random forest + graph features	90.7%	80.6%	0.85	0.92	0.22%
Gradient boosting + graph features ★	94.8%	83.8%	0.89	0.94	0.12%

848,752 synthetic Bangladesh-style MFS transactions with seven planted typologies and legitimate look-alikes. Trained on 2 networks, tuned on 1, tested on 3 unseen networks with different seeds (mean shown). Labels are exact by construction, so treat these as upper bounds for real upay data.

What the network catches that transaction rules miss

3 HELD-OUT NETWORKS · 1,010 SUSPICIOUS ACCOUNTS

Transaction rules only 324 of 1,010 suspicious accounts caught



Cygnus (graph + ML) 846 of 1,010 suspicious accounts caught



561 missed by rules, caught via the network

Of the 686 accounts the rules miss, the graph model catches 561. On evasive variants built to stay under rule thresholds, rules catch 9.1%, the model without graph features 66.1%, with graph features 82.6%. False alarms fall from 904 to 46.

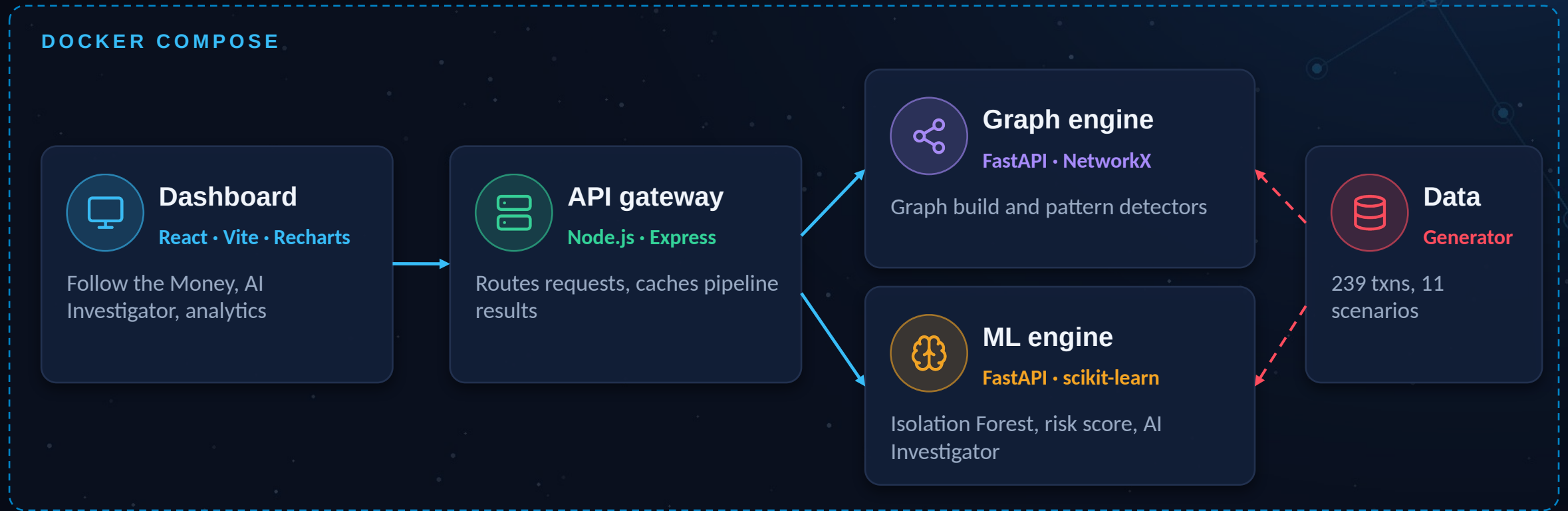
FEATURE IMPORTANCE · PR-AUC DROP WHEN SHUFFLED



Adding feature families, PR-AUC: transactional 0.66 → + temporal 0.82 → + graph and flow tracing 0.93 → + 2-hop neighbourhood 0.94

● Graph / network ● Transactional

Four services, one Docker Compose stack



React 19

Vite

Tailwind

Recharts

Node.js

Express

Python

FastAPI

NetworkX

pandas

scikit-learn

pytest

Docker

Measured today, with a clear path to upay volume

MEASURED ON A LAPTOP TODAY

1,061,510

transactions scored in 18.9 s

~56,000 / s

scoring throughput, one process

< 6 ms

API response time, p95 (every endpoint)

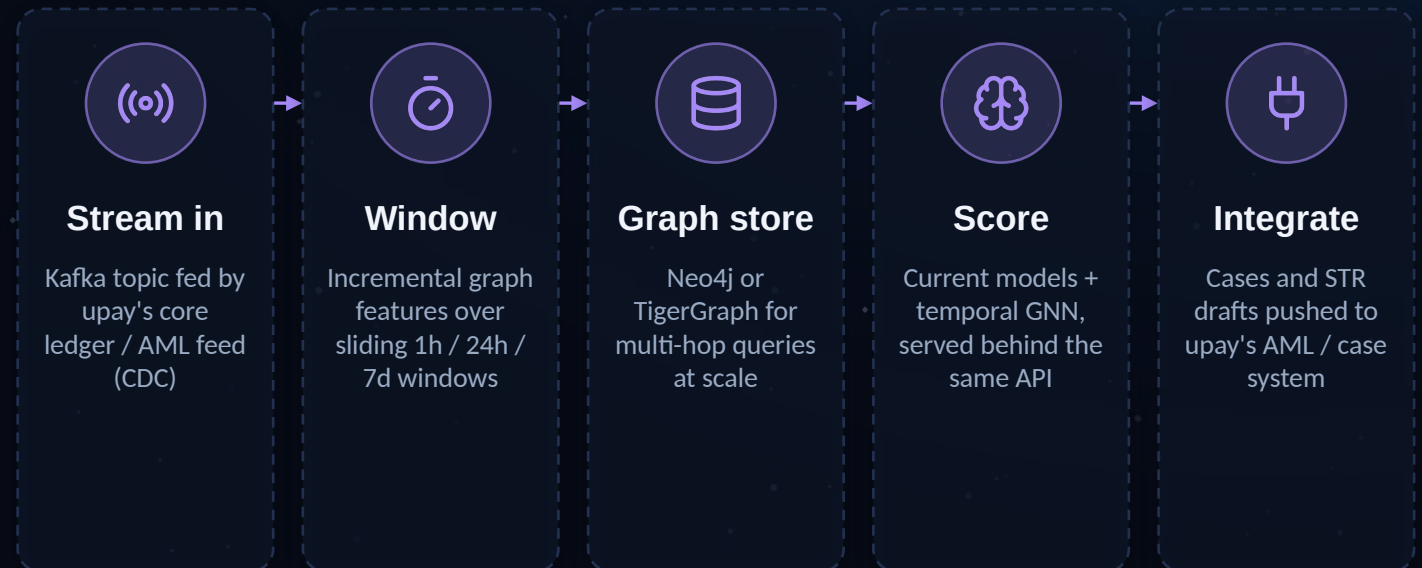
0.21 s

re-score the last 24 hours (9,508 tx)

8-core laptop, Python 3.11, one process in Docker, no GPU, peak 1.9 GB.
Full pipeline with graph and evidence text: ~8,600 tx/s.

upay-scale target All Bangladesh MFS together average ~250 transfers a second [2]. Our batch scoring already runs ~56,000 a second on one laptop process; the streaming layers below make it real-time for upay.

TARGET ARCHITECTURE · ROADMAP



Safeguards built today, and what production adds

BUILT AND TESTED IN THE PROTOTYPE

-  **Strict input validation** bad amounts, NaN/Inf, bad timestamps, self-transfers, duplicates rejected
-  **Role-based access** sign-in as AML analyst or compliance officer
-  **Hash-chained audit log** case actions and data reveals are chained, so tampering shows
-  **PII masking** personal details masked by default; every reveal is logged
-  **Bias check** legit agents flagged 98.7% (first prototype) → 0.57% now; no district effect ($p = 0.30$)
-  **Grounded explanations** every sentence quotes a computed metric; the optional LLM only rephrases
-  **Human in the loop** scores prioritise review; analysts decide and file STRs

PRODUCTION ROADMAP

- ▶ SSO with upay identity and MFA for analysts
- ▶ TLS everywhere and encryption at rest for case data
- ▶ Write-once audit storage and regular external penetration tests
- ▶ Model-drift and alert-volume monitoring with monthly threshold review
- ▶ Privacy governance: data minimisation, retention limits, access reviews
- ▶ Fairness review on real data across regions, account types and agent tiers
- ▶ Cross-provider analytics without sharing raw PII (federated learning or secure aggregation)

What it saves an AML team, and how a pilot proves it

73.6%

fewer alerts at equal recall

324 alerts instead of 1,228 to find the same 324 accounts

904 → 46

false alarms

on 39,490 held-out accounts

32% → 84%

suspicious accounts caught

rules vs graph model, same networks

76 → 21 min

analyst time per true case

at 20 minutes per alert (assumed)

MODELLLED SAVINGS · 20 MIN PER ALERT, ASSUMED

Assumption: 20 minutes of analyst review per alert. The pilot replaces it with upay's measured time.

Rules: 1,228 alerts = 409 analyst hours, finding 324 suspicious accounts.

Graph model: 892 alerts = 297 analyst hours, finding 846.

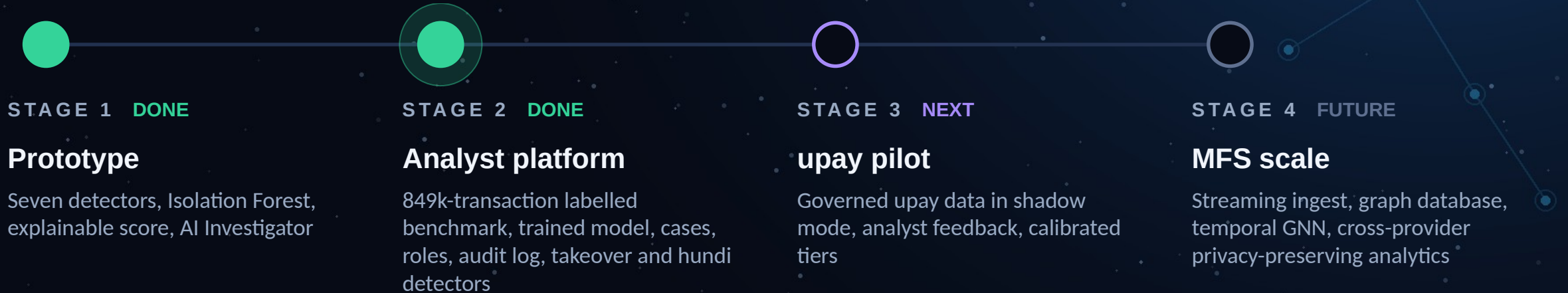
Same 324 accounts with the graph model: 324 alerts, 108 hours instead of 409.

Synthetic networks are ~2.5% suspicious, far above production, so real precision will be lower.

PROPOSED 8-WEEK PILOT WITH UPAY

- ▶ Shadow mode on governed upay data: Cygnus scores, analysts work as usual
- ▶ Analysts label every alert; we measure precision@100 and recall on confirmed cases
- ▶ Time each case with and without Cygnus to get real minutes saved
- ▶ Calibrate thresholds and tiers with upay's AML team
- ▶ Exit report: ROI, false-positive cost and a go / no-go for production

From prototype to a upay pilot to MFS scale



MFS providers like upay

Spot mule networks before funds are cashed out.



Compliance teams

Trace a multi-hop flow in seconds instead of hours.



Account holders

Fewer recruited mules and victims, more trust in digital money.

Don't just flag transactions.
Follow the money.

CYGNUS AI

Graph intelligence and explainable AI for safer mobile money
in Bangladesh.

Thank you

Built for upay • AI Dev Fest 2026

github.com/ManamiMayoki/CPC_DIU_HACKATHON_2026

